

1.To write a Python program to add two numbers entered by the user.

```
1 a = int(input("Enter first number:"))
2 b = int(input("Enter second number:"))
3 sum = a + b
4 print("Sum=", sum)
```

```
Enter first number:1
Enter second number:3
Sum= 4
```

2.To write a Python program to interchange the values of two variables.

```
1 a = int(input("Enter first number:"))
2 b = int(input("Enter second number:"))
3 print("Before Swapping")
4 print("a=", a)
5 print("b=", b)
6 temp =a
7 a=b
8 b=temp
9 print("After Swapping")
10 print("a=", a)
11 print("b=", b)
```

```
Enter first number:6
Enter second number:9
Before Swapping
a= 6
b= 9
After Swapping
a= 9
b= 6
```

3.To write a python program to find the largest among three numbers using conditional statements.

```
1 a = int(input("Enter first number"))
2 b = int(input("Enter second number"))
3 c = int(input("Enter third number"))
4 if a>b and a>c:
5     print("Largest=", a)
6 elif b>c:
7     print("Largest=", b)
8 else :
9     print("Largest=", c)
```

```
Enter first number5
Enter second number7
```

```
Enter third number10  
Largest= 10
```

4.To write a python program to check whether a given number is even or odd.

```
1 num = int(input("Enter a number:"))  
2 if num%2==0:  
3     print("Even:", num)  
4 else:  
5     print("Odd:", num)
```

```
Enter a number:6  
Even: 6
```

5.To write a Python program to determine whether a number is positive, negative or zero.

```
1 num = int(input("Enter a number:"))  
2 if num > 0:  
3     print("Positive:",num)  
4 elif num < 0:  
5     print("Negative:",num)  
6 else:  
7     print("Zero:",num)
```

```
Enter a number:89  
Positive: 89
```

6.To write a Python program to check whether a given year is leap year.

```
1 year = int(input("Enter year:"))  
2 if year % 4 == 0:  
3     print("Leap Year")  
4 else:  
5     print("Not Leap Year")
```

```
Enter year:2009  
Not Leap Year
```

7.To write a Python Program to perform basic arithmetic operations using operators.

```
1 a = int(input("Enter first number:"))  
2 b= int(input("Enter second number:"))  
3  
4 print("Addition:", a + b)  
5 print("Subtraction:", a - b)  
6 print("Multiplication:", a * b)  
7 print("Division:", a / b)
```

```
Enter first number:0
Enter second number:8
Addition: 8
Subtraction: -8
Multiplication: 0
Division: 0.0
```

8.To write a Python program to perform basic multiplication table of a given number.

```
1 num = int(input("Enter number:"))
2 for i in range(1,11):
3     print(num, "x", i,"=",num*i)
```

```
Enter number:9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
```

9.To write a Python program to calculate the factorial of a number using a loop.

```
1 n = int(input("Enter number:"))
2 fact = 1
3 for i in range(1,n + 1):
4     fact= fact*i
5 print("Factorial=", fact)
```

```
Enter number:4
Factorial= 24
```

10.To write a Python program to find the sum of the first N natural numbers.

```
1 n = int(input("Enter N:"))
2 sum=0
3 for i in range(1,n + 1):
4     sum = sum+i
5 print("Sum=", sum)
```

```
Enter N:6
Sum= 21
```

11.To write a Python program to print numbers form 1 to N using a for loop.

```
1 n = int(input("Enter N:"))
2 for i in range(1,n+1):
```

```
3 print(i)
```

Enter N:6

```
1
2
3
4
5
6
```

12.To write a Python program to print numbers in reverse order from N to 1.

```
1 n = int(input("Enter number:"))
2 while n >= 1:
3     print(n)
4     n=n-1
```

Enter number:7

```
7
6
5
4
3
2
1
```

13.To write a Python program to print all even numbers up to a given limit.

```
1 n = int(input("Enter limit:"))
2 for i in range(2,n+1,2):
3     print(i)
4
```

Enter limit:8

```
2
4
6
8
```

14.To write a Python program to reverse the digits of a given number.

```
1 n = int(input("Enter a number:"))
2 reversed_num = 0
3 original_num = n
4 while n > 0:
5     digit = n % 10
6     reversed_num = reversed_num * 10 + digit
7     n = n // 10
8 print(f"Original number: {original_num}")
9 print(f"Reversed number: {reversed_num}")
```

Enter a number:679

Original number: 679

Reversed number: 976

15. To write a Python program to count the number of digits in a given integer.

```
1 n = (input("Enter number:"))
2 print("Total digit=", len(n))
```

Enter number:5667
Total digit= 4

16.To write a Python program to calculate the sum of digits of a given number.

```
1 n = input("Enter a number: ")
2 sum = 0
3 for i in n:
4     sum = sum + int(i)
5 print("Sum =", sum)
```

Enter a number: 76
Sum = 13

17. To write a Python program to check whether a given number is prime.

```
1 n = int(input("Enter a number: "))
2 if n == 2:
3     print("Prime Number")
4 elif n % 2 == 0:
5     print("Not Prime Number")
6 else:
7     print("Prime Number")
```

Enter a number: 8
Not Prime Number

18. To write a Python program to generate the Fibonacci series up to N terms.

```
1 n = int(input("Enter number of terms: "))
2 a = 0
3 b = 1
4 for i in range(n):
5     print(a)
6     c = a + b
7     a = b
8     b = c
```

Enter number of terms: 2
0
1

19. To write a Python program to check whether a string is a palindrome.

```
1 s = input("Enter a string: ")
2 rev_s = s[::-1]
3 if s == rev_s:
4     print("The string is a palindrome.")
5 else:
6     print("The string is not a palindrome.")
```

Enter a string: 879
The string is not a palindrome.

20. To write a Python program to check whether a given number is an Armstrong number.

```
1 n = int(input("Enter a number: "))
2 temp = n
3 sum = 0
4 while temp > 0:
5     digit = temp % 10
6     sum = sum + digit * digit * digit
7     temp = temp // 10
8 if sum == n:
9     print("Armstrong Number")
10 else:
11     print("Not Armstrong Number")
```